Discrete-time signal synthesis using the IDFT

Report draft - repository URL pending publication.

1. Objective and input

Synthesize a discrete-time signal from X = [6, 2, 4, 3, 4, 4, 0, 0, 0, 0]^T. Use N = 10 and illustrative additional lengths N = 16 and N = 20, appending zeros to the input spectrum.

2. Matrix method

Define W[n,k] = exp(-j 2 pi n k/N), for n,k = 0,...,N-1. Define K[n,k] = exp(+j 2 pi n k/N)/N. Thus K = W^H/N = W^(-1), x = K X and K W = I. The letter conventions are chosen explicitly because the exercise does not specify them. The executed notebook and its PDF export print complete W and K matrices for each N.

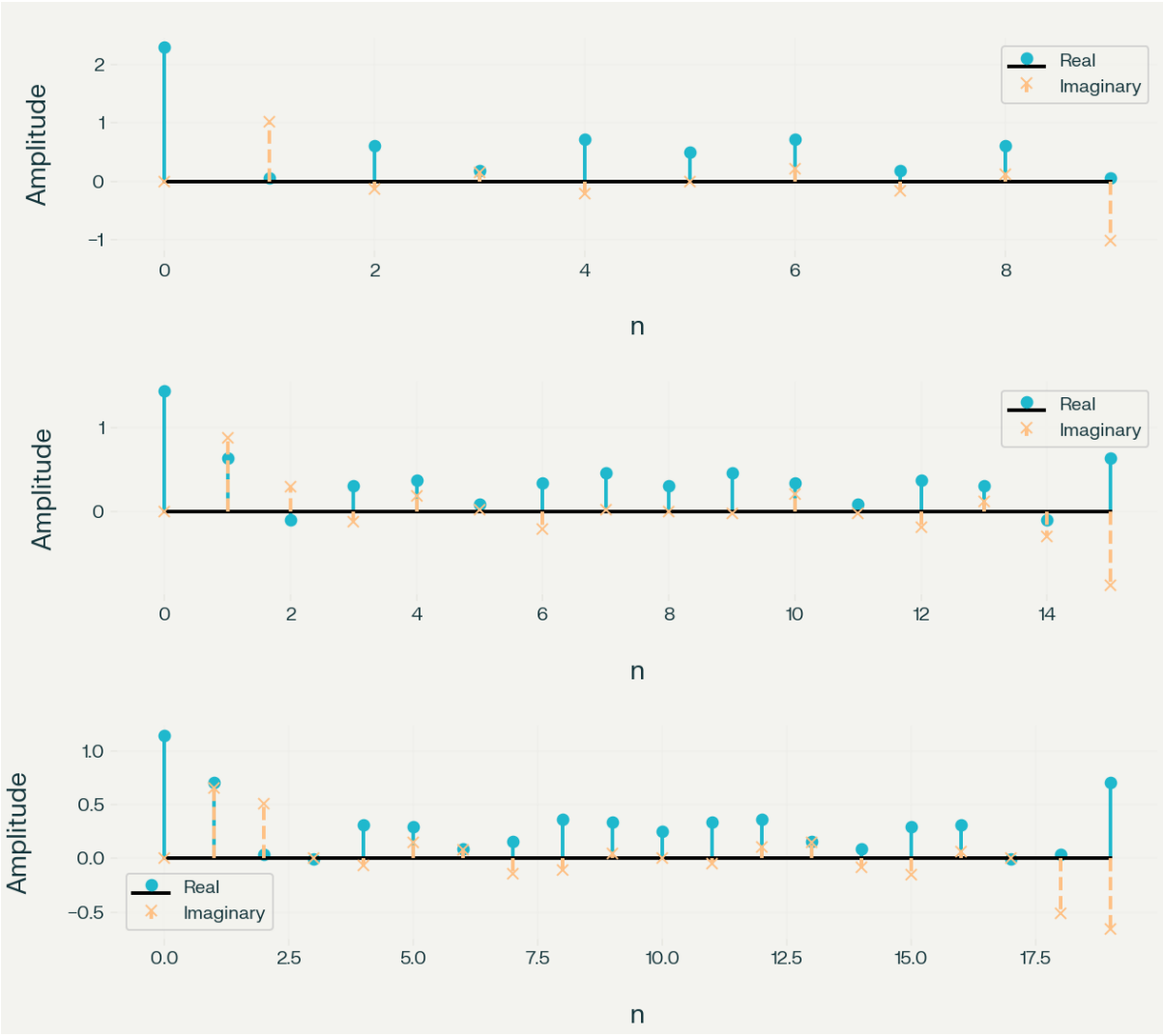
3. Synthesis formula

x[n] = (6 + 2 exp(j 2 pi n/N) + 4 exp(j 4 pi n/N) + 3 exp(j 6 pi n/N) + 4 exp(j 8 pi n/N) + 4 exp(j 10 pi n/N))/N.

4. Numerical result for N = 10

```
n= 0:  2.300000 +0.000000j
n= 1:  0.069098 +1.018411j
n= 2:  0.619098 -0.131433j
n= 3:  0.180902 +0.159184j
n= 4:  0.730902 -0.212663j
n= 5:  0.500000 -0.000000j
n= 6:  0.730902 +0.212663j
n= 7:  0.180902 -0.159184j
n= 8:  0.619098 +0.131433j
n= 9:  0.069098 -1.018411j
```

5. Synthesized signals



6. Interpretation and verification

The output is complex because the input spectrum is not conjugate symmetric. At n = 0 the values are 23/N: 2.3, 1.4375 and 1.15 for N = 10, 16 and 20. Keeping the coefficients fixed while increasing N changes their normalized frequencies and the 1/N scale; this is not amplitude-preserving interpolation of an unchanged waveform.

The notebook checks both K W = I and agreement with numpy.fft.ifft for every N.

7. Repository and submission

Repository URL: PENDING. Publish the notebook and its PDF export, then replace this placeholder with the actual repository link before submitting this report to eLearning. This draft has not been uploaded or submitted.

Reference: Brian McFee, Digital Signals Theory, section 7.2.
https://brianmcfee.net/dstbook-site/content/ch07-inverse-dft/InverseDFT.html